Should the Fusion Weight Change with the Question?

Part 3: Do Better Frames Give Better Answers? (Q4)

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Team and my part

Member	Main contribution	
Jianxiang He	Problem and setup; score cache and data pipeline; Q1 analysis; overall study (Q1–Q4)	
Longtao Jiang	Continual learning experiments and analysis (Q2–Q3)	
Makhmud Assentay	**Downstream question-answering evaluation (Q4)**	**this talk**

Background: two clues, one weight

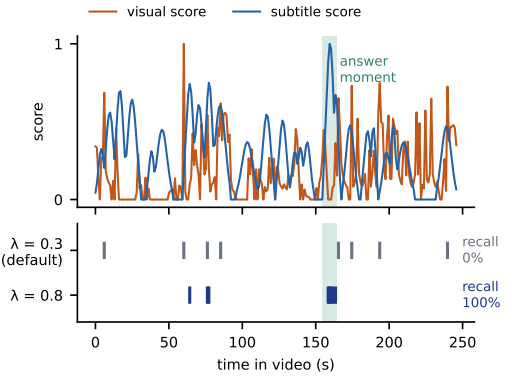
A VLM can only look at **about 8 frames** of a long video, so we must pick them well.

VSI scores every second with two clues:

- **Visual**: do the question's objects appear?
- **Subtitle**: does the subtitle match the question?

$$S(t) = (1 - \lambda)\,\text{visual}(t) + \lambda\,\text{subtitle}(t)$$

and keeps the top 8 seconds. VSI uses $\lambda = 0.3$ **for every question**.



Real example: $\lambda = 0.3$ **misses** the answer moment (green), $\lambda = 0.8$ **hits** it.

My part: does it help answers?

- Our team changed λ to pick frames that **hit the answer moment** more often (recall@8).
- But the real goal is **answering the question**.
- Recall@8 is only a stand-in: does it predict accuracy?

My part (Q4)

Built and ran the question-answering evaluation: every frame choice is given to a real VLM, and accuracy is compared on the same questions.

How we test answers

- **Model**: Qwen3-VL-8B-Instruct
- **Input**: the 8 chosen frames, each with the subtitle shown at that moment, then the multiple-choice question
- **Output**: one option letter
- **Frame choices**: even spacing, $\lambda = 0, 0.3, 0.7, 1$, oracle λ, learned predictors
- 753 questions; learned predictors on their own test questions (3×229), **paired** with the default
- Same frames ⇒ one model call: 3,482 calls in total

Prompt format

```
Frame 1 (12s):  [image]
Subtitle:  ...
...
Frame 8 (231s):  [image]
Question:  ...
A. ...  B. ...  C. ...
Answer with the option's letter from the
given choices directly.
```

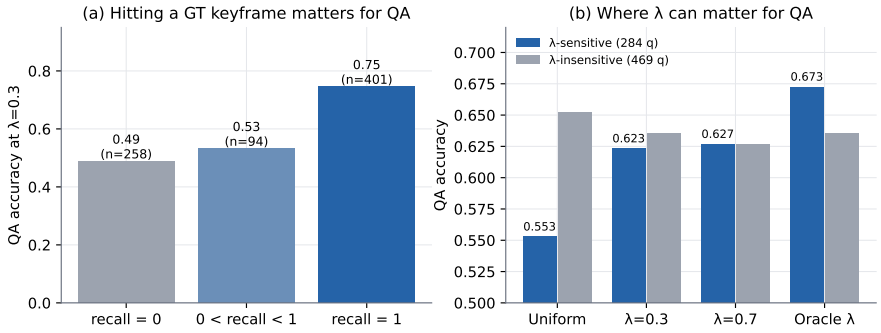
Result: same accuracy as default

Frames chosen by	QA accuracy
Even spacing	0.615
$\lambda = 0$ (picture only)	0.604
$\lambda = 0.3$ (default)	0.631
$\lambda = 0.7$ (best recall)	0.627
$\lambda = 1$ (subtitles only)	0.615
Oracle λ	0.649
LwF predictor (paired)	$+0.4$

95% CI $[-1.5, +2.3]$, McNemar $p = 0.77$

- Learned λ = default (**+0.4**, not significant)
- Even the oracle adds only **1.8**
- $\lambda = 0.7$ has the best recall but answers slightly **worse**
- Using both clues beats either one alone

Frames matter, for some questions



(a) Accuracy **0.49** when the 8 frames hit no answer moment vs. **0.75** when they hit all.
(b) On the **284** questions where λ changes the frames, the oracle λ answers **5 points** more; elsewhere nothing changes.

Why recall is not accuracy

Reasons

1. Only **38%** of questions react to λ at all.
2. Recall@8 is coarse: about 2 answer moments per question.
3. The VLM often answers from subtitles or common sense (even spacing: 0.615).

On the 284 sensitive questions

LwF predictor: **+1.5** over the default, but 95% CI $[-2.3, +5.6]$ includes 0. It captures only a small part of the 5-point room.

Takeaways from Q4

1. On all questions, a learned weight gives **the same answers** as the default.
2. Frames still matter: hitting the answer moment raises accuracy from 0.49 to 0.75.
3. The weight matters only for the **38%** of questions where it changes the frames; there the best weight adds 5 points.
4. Lesson: report **answer accuracy**, not only recall, and look at the questions where the change actually applies.

Thank You

Questions?

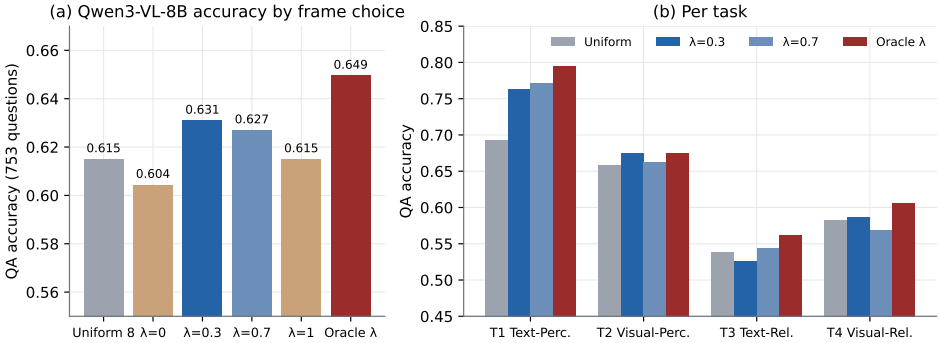
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Backup: accuracy per type

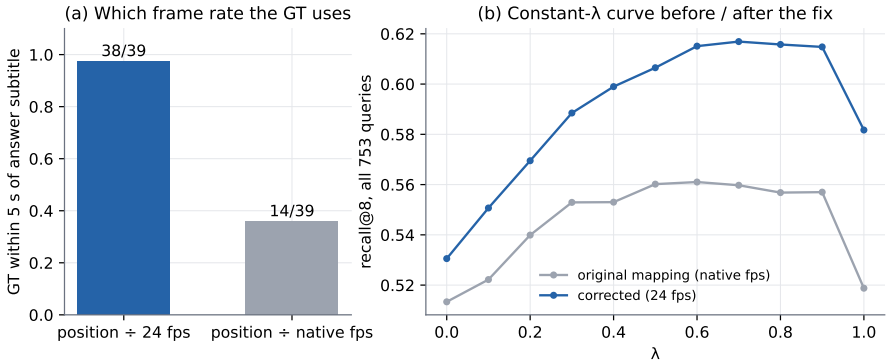


	T1 text perc.	T2 visual perc.	T3 text rel.	T4 visual rel.
$\lambda = 0.3$	0.764	0.675	0.526	0.587
Oracle λ	0.795	0.675	0.561	0.606

Backup: significance tests

- Each predictor is tested on the 229 test questions of its own split, so it never saw them in training.
- **Paired**: on exactly the same questions as the default $\lambda = 0.3$.
- LwF vs. default over 3×229 questions: 24 questions only LwF got right, 21 only the default got right.
- **McNemar test** on these disagreements: $p = 0.77$. **Bootstrap** 95% CI of the difference: $[-1.5, +2.3]$ points.
- Details: bf16, at most 448×448 pixels per frame, greedy decoding, subtitles within ± 1 s of each frame.

Backup: annotation frame rate



LongVideoBench counts annotated frames at a fixed 24 fps, not the real 30 fps. (a) 38/39 annotations match the answer subtitle at 24 fps, only 14/39 at the real rate. (b) Recall vs. λ before and after the fix.

Backup: recall vs. λ

